

H Mohammed Yusuf | B.Tech ECE Student | Robotics & Embedded Systems Enthusiast

National Institute of Technology Puducherry (NITPY) – Karaikal, India

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High-achieving ECE Sophomore (CGPA 9.24) at NIT Puducherry with practical expertise in Industrial Robotics and Embedded Systems. Project Intern at EXOR Robotics and Co-Creator of the YUJAI Development Board. Proficient in DSP, Analog Communication, and IoT frameworks (ESP32). Proven leader as Maths Subject Head at LEAP NITPY. Seeking to engineer innovative hardware solutions that bridge theory and application.

Education

National Institute of Technology Puducherry (NITPY) <i>Bachelor of Technology in Electronics and Communication Engineering, CGPA: 9.24/10.0</i> Currently in 2nd Year. Consistently ranked among top performers in the batch.	Karaikal 2024–2028
Aditya Vidhyashram Residential School <i>Higher Secondary Education (Class XII), Percentage: 91.2%</i> Specialized in Physics, Chemistry, and Mathematics.	Puducherry 2022–2024
Aditya Vidhyashram Residential School <i>Secondary Education (Class X), Percentage: 92.8%</i> Achieved academic excellence in Science and Mathematics.	Puducherry 2022

Experience

EXOR ROBOTICS <i>Project Intern</i> ○ Gained hands-on exposure to industrial robotics and automation systems. ○ Collaborated with the engineering team to test and troubleshoot hardware prototypes. ○ Applied theoretical concepts of sensors and actuators in a real-world manufacturing environment.	Puducherry <i>Internship</i>
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Key Projects

YUJAI 8051 Development Board <i>Co-Architect & Hardware Developer</i> ○ Designing a custom, cost-effective 8051 Microcontroller Board to simplify embedded learning. ○ Tech Stack: 8051 Architecture, PCB Design, Embedded C. ○ Engineered the power supply unit and I/O interfacing circuits for stability and modularity.	NIT Puducherry <i>Ongoing</i>
Smart Blind Stick (Obstacle Detection & GPS) <i>Team Lead & Developer</i> ○ Developing an assistive device with ultrasonic sensors for obstacle detection and GPS/GSM for emergency SOS navigation. ○ Integrating IoT modules to provide real-time location tracking for caregivers.	Smart India Hackathon <i>SIH 2025</i>
Voice Controlled Car using ESP32 <i>Developer</i>	Academic Project <i>IoT & Robotics</i>

- Built a WiFi-controlled robot car utilizing the ESP32 module, interfaced with L298N motor drivers.
- Developed a mobile interface to send voice commands, processed via speech-to-text API for navigation.

Technical Prototypes & Mini Projects

- **Audio Amplifier System (LM386):** Designed and fabricated a low-power audio amplifier circuit with gain control and noise filtering.
- **Sensor Data Logging in MATLAB:** Interfaced sensors with microcontroller UART to log real-time data into MATLAB for signal visualization and analysis.
- **Pocket Dashboard (ESP32):** Created a portable web-server based dashboard to monitor sensor readings remotely.
- **Line Follower Bot:** Implemented an autonomous robot using IR sensor arrays and PID control logic for precise path tracking.

Technical Skills

Core Domain: Digital Signal Processing (DSP), Analog Communication, Digital Electronics

Hardware: Microcontrollers (8051, ESP32, Arduino), Sensors, Actuators, PCB Design

Programming: C, C++, Python, Embedded C, MATLAB

Tools: Keil μ Vision, LTSpice, KiCad/EasyEDA, Arduino IDE

Soft Skills: Leadership, Mentoring, Strategic Planning, Problem Solving

Positions of Responsibility

LEAP NITPY

Maths Subject Head

Karaikal

2024–Present

- Leading peer-learning initiatives and organizing mathematics doubt-solving sessions for first-year students.
- Mentoring students to improve their academic performance and conceptual understanding.

SAE NITPY (Society of Automotive Engineers)

Member, Power Train Team

Karaikal

2024–Present

- Contributing to the design and analysis of power transmission systems for student formula vehicles.
- Researching efficient energy management strategies for automotive applications.

Achievements

- **Inspire Awards 2021:** Recognized for developing a "Social Distancing Bot" during the pandemic, showcasing early innovation in robotics.
- Maintained a consistent **CGPA of 9.24** in B.Tech ECE.
- Active participant in Smart India Hackathon (SIH) 2025.

Languages

English: Professional Proficiency

Tamil: Native Proficiency

Hindi: Working Proficiency